



United International University
Department of Computer Science and Engineering, Spring 26
CSE 3812 - Artificial Intelligence Laboratory
Mid Exam - Total Marks: 20 - Duration: 50 min

Question 1

3

A library system stores book information in a dictionary. You have:

```
patient = {  
    "name": "Ayesha",  
    "age": 30,  
    "disease": "Flu",  
    "admitted": True  
}
```

Task: The patient has been discharged. Write code to:

- update "admitted" to False
- add a new key "discharge_date": "2026-05-01".
- Print the final dictionary.

Question 2

2

What will be the output of the following code?

```
t = (10, 20, 30, 40, 50)  
print(t[1:4])  
print(t[::-1])  
print(t[::2])
```

Question 3

3

Complete the program to find all unique elements from the list and store their square values in a dictionary.

```
numbers = [1, 2, 3, 2, 4, 3, 5]  
squares = _____  
for num in _____:  
    unique = _____(_____)  
for item in _____:  
    squares[_____] = _____ ** 2  
print(_____)
```

Question 4

4

A university system tracks student records. Each student has a name and a list of marks. Complete the class so the main code runs correctly and produces the expected output.

```

class Student:
    # TODO 1: Complete the constructor
    # Takes name and marks (a list) as parameters
    def __init__(_____, name, marks):
        _____ = _____
        _____ = _____

    # TODO 2: Complete this method
    # It should return the average of marks
    def average(self):
        return _____ / _____
    def display(self):
        print(f"{self.name} | Average: {self.average():.1f}")

# Main Code - DO NOT MODIFY
s1 = Student("Nadia", [80, 90, 75, 85])
s2 = Student("Imran", [60, 70, 65, 55])
s1.display()
s2.display()

```

Expected Output:

```

Nadia | Average: 82.5
Imran | Average: 62.5

```

Question 5

8

A delivery drone is flying across a city grid to deliver a package from Start Point S(0,0) to Destination D(5,5). The drone can only move Up, Down, Left, and Right (no diagonal moves). The city grid has several blocked buildings in between. The navigation team is debating which heuristic function to use in their A* Search Algorithms. Two engineers propose different heuristics:

- Engineer A proposes: "Let's just use $h(n) = 0$ for every node"
- Engineer B proposes: "Let's use $h(n) = \text{straight line distance} \times 10$ even if actual path is longer"
- Engineer C proposes: "Let's use Manhattan Distance as our heuristic"

Based on this scenario, answer the following:

- Explain why Engineer A's heuristic $h(n) = 0$ is considered a bad heuristic even though A* will still find the correct path. What well-known algorithm does A* become when $h(n) = 0$? [2 Marks]
- Explain why Engineer B's heuristic is also considered dangerous and bad. What problem occurs when a heuristic overestimates the actual cost? Use the term admissibility in your answer. [2 Marks]
- Explain why Engineer C's Manhattan Distance is the best heuristic for this drone scenario. Use the formula below and show a sample calculation from S(0,0) to D(5,5): [2 Marks]

Three engineers have proposed three different heuristics for the drone navigation system. What is your opinion? Which engineer proposed the best solution and why? [2 Marks]



United International University
Department of Computer Science and Engineering, Spring 26
CSE 3812 - Artificial Intelligence Laboratory
Mid Exam - Total Marks: 20 - Duration: 50 min

Question 1

3

A library system stores book information in a dictionary. You have:

```
book = {  
    "title": "Python",  
    "author": "Messi",  
    "year": 2020,  
    "available": True  
}
```

Task: A user borrows the book. Write code to

- update "available" to False
- add a new key "borrower": "Rahim".
- Print the final dictionary.

Question 2

2

What will be the output of the following code?

```
d = {"a": 1, "b": 2, "c": 3}  
d["a"] = 10  
d["d"] = 4  
print(len(d))
```

Question 3

3

Complete the program to count the frequency of each character in the string "programming" using a dictionary. - 3 Marks

```
text = "programming"  
freq = {}  
for char in text:  
    if char ____ freq:  
        freq[char] ____ 1  
    else:  
        freq[char] = ____  
print(____)
```

Question 4

4

A dog shelter system stores each dog's name and breed. Complete the class so the main code runs correctly.

```
class Dog:
    # TODO 1: Complete the constructor
    def __init__(_____, name, breed):
        _____ = _____
        _____ = _____

    # TODO 2: Complete this method
    def bark(self):
        print(f"_____ is _____. It is barking!")

# Main Code - DO NOT MODIFY
d1 = Dog("Rex", "Labrador")
d1.bark()
```

Expected Output:

Rex is Labrador. It is barking!

Question 5

8

A delivery drone is flying across a city grid to deliver a package from Start Point S(0,0) to Destination D(5,5). The drone can only move Up, Down, Left, and Right (no diagonal moves). The city grid has several blocked buildings in between. The navigation team is debating which heuristic function to use in their A* Search Algorithms. Two engineers propose different heuristics:

- Engineer A proposes: "Let's just use $h(n) = 0$ for every node"
- Engineer B proposes: "Let's use $h(n) = \text{straight line distance} \times 10$ even if actual path is longer"
- Engineer C proposes: "Let's use Manhattan Distance as our heuristic"

Based on this scenario, answer the following:

- a) Explain why Engineer A's heuristic $h(n) = 0$ is considered a bad heuristic even though A* will still find the correct path. What well-known algorithm does A* become when $h(n) = 0$? [2 Marks]
- b) Explain why Engineer B's heuristic is also considered dangerous and bad. What problem occurs when a heuristic overestimates the actual cost? Use the term admissibility in your answer. [2 Marks]
- c) Explain why Engineer C's Manhattan Distance is the best heuristic for this drone scenario. Use the formula below and show a sample calculation from S(0,0) to D(5,5): [2 Marks]

Three engineers have proposed three different heuristics for the drone navigation system. What is your opinion? Which engineer proposed the best solution and why? [2 Marks]